Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

PARTICIPANT DETAILS

a. Participant name : **Hariharan C**

b. Problem Statement : **Build and deploy a production-ready AI agent using Gemini and ADK that performs a single task via an HTTP endpoint on Cloud Run.**

c. Github repo link : ***https://github.com/BlackWizard123/Track-1---Financial_Risk_Classifier_Agent***

d. Cloud Run URL : ***https://financial-risk-agent-1059652519537.asia-south1.run.app/***

Brief about the idea

We built an **AI-powered Financial Risk Classifier Agent** that analyzes a user's financial situation from natural language input.

- Takes unstructured financial text (income, expenses, debts)
- Uses **Gemini** to understand and evaluate financial signals
- Classifies risk into **Low / Medium / High**
- Returns **structured insights** with:
    - Reasoning
    - Summary
    - Improvement suggestions
    - Stability plan

Deployed as a **scalable API on Cloud Run**, making it easily accessible and production-ready.

Meeting the Build Criteria

How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

- Implemented a **single AI agent** focused on financial risk classification
- Integrated **ADK structure** to align with agent-based architecture
- Designed a clear **HTTP endpoint (/analyze)** using FastAPI
- Ensured **input → processing → structured output** flow
- Used **Gemini model** for intelligent analysis and reasoning
- Deployed on **Cloud Run** for scalability and accessibility

✓ Fully satisfies the requirement of building a **production-ready, callable AI agent**

Opportunities

- **How different is it from any of the other existing ideas?**

Uses natural language input instead of structured financial data, making it more intuitive than traditional tools

- **How will it be able to solve the problem?**

Translates user financial descriptions into clear risk insights and actionable recommendations

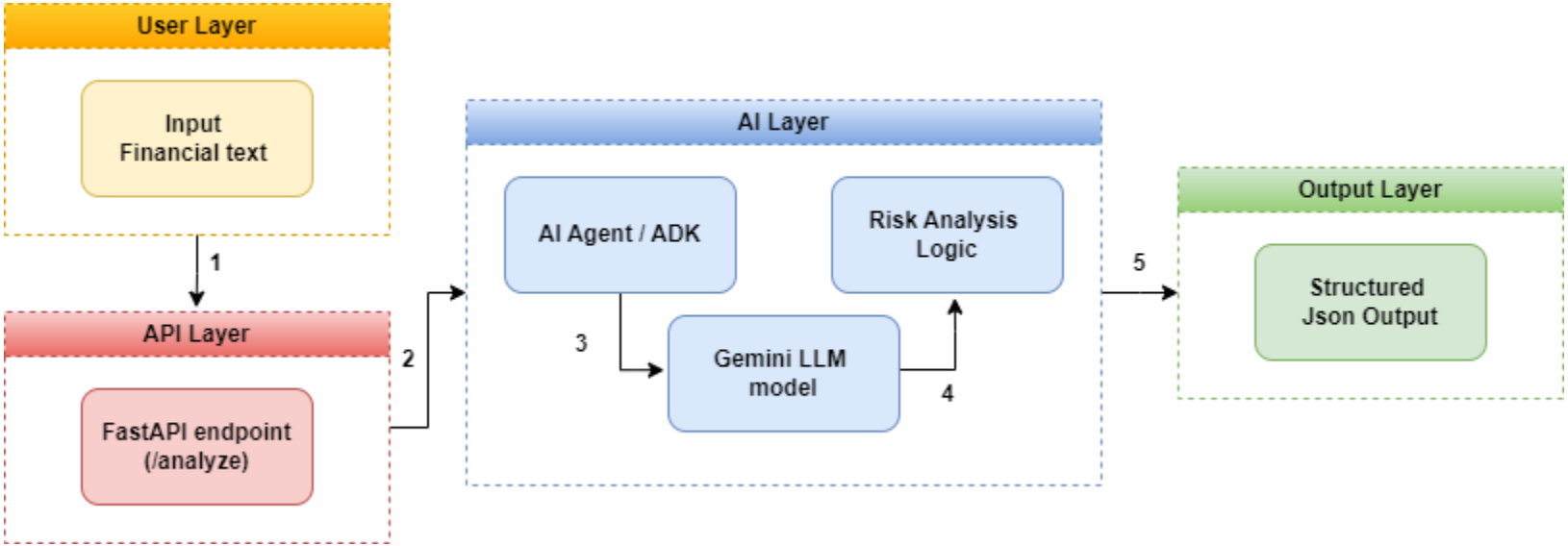
- **USP of the proposed solution**

Translates user financial descriptions into clear risk insights and actionable recommendations

List of features offered by the solution

- AI-powered financial risk classification (Low / Medium / High)
- Accepts natural language financial input
- Structured JSON output (not just plain text)
- Clear reasoning behind risk classification
- Actionable improvement suggestions
- Stability planning guidance
- FastAPI-based HTTP endpoint (/analyze)
- Deployed on Cloud Run (scalable & accessible)
- ADK-based agent structure
- Optional UI for interactive demo

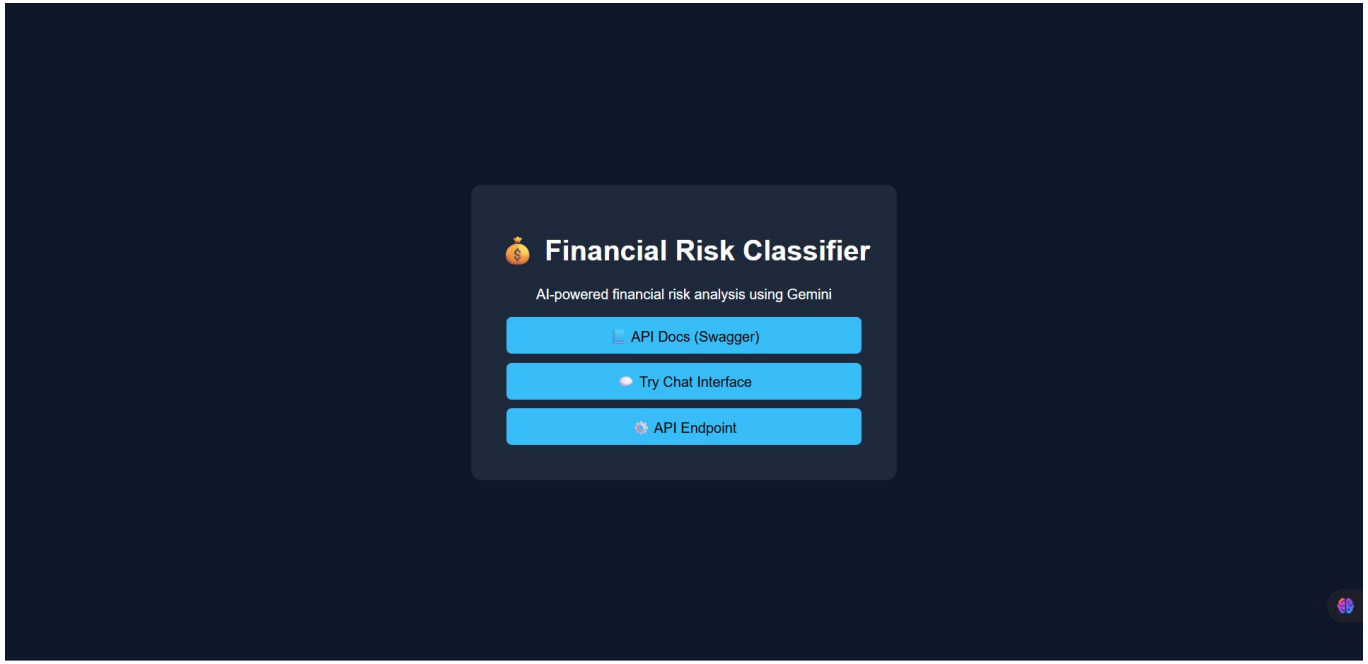
Process flow diagram or Use-case diagram



Technologies used in the solution

- **Python** – Core programming language
- **FastAPI** – Backend API framework
- **Google Gemini** – AI model for analysis and reasoning
- **Docker** – Containerization
- **Google Cloud Run** – Serverless deployment
- **Google ADK (Agent Development Kit)** – Agent structure and orchestration
- **HTML/CSS (Optional UI)** – Simple user interface for demo

Snapshots of the prototype



https://financial-risk-agent-1059652519537.asia-south1.run.app/

💰 Financial Risk Analyzer

YOU :
I have a stable government job with a monthly salary of ₹1.2 Lakhs. My total monthly expenses, including rent and bills, are ₹40,000. I have an emergency fund covering 12 months of expenses in a high-interest savings account. My only debt is a small car loan with a 4% interest rate. Evaluate my overall financial risk level and suggest if my emergency fund is 'over-capitalized'.

AI RESPONSE :
📊 Risk Level: Low

🧠 Summary:
Excellent financial health characterized by a stable income, low expenditure, substantial savings, and minimal, manageable debt.

⚠️ Key Reason:
The user has a stable government job with a high monthly income of ₹1.2 Lakhs, significantly exceeding their low monthly expenses of ₹40,000. They have a robust emergency fund covering 12 months of expenses and minimal debt, limited to a small car loan.

✅ Improvements:
• **Evaluate Emergency Fund Allocation:** While highly secure, an emergency fund covering 12 months of expenses might be 'over-capitalized' given a stable government job. Consider if 6-9 months would suffice, potentially freeing up excess capital (e.g., ₹1.2 - ₹2.4 Lakhs) for higher-return investments.
• **Establish a Diversified Investment Portfolio:** With significant disposable income (₹80,000/month) and an adequate emergency fund, focus on long-term wealth creation through diversified investments (e.g., equity mutual funds, index funds, real estate) aligned with personal financial goals.
• **Optimize Debt Strategy (Optional):** Although the car loan is small and at a low interest rate, compare its 4% interest with potential investment returns. If investment returns are consistently higher, prioritize investing; if not, consider early repayment for peace of mind.

📈 Stability Plan:
• **Maintain High Savings & Investment Rate:** Continue to regularly save and invest the substantial surplus (₹80,000 per month) after covering expenses.
• **Regular Financial Review:** Periodically review income, expenses, and investment performance to ensure alignment with financial goals and adjust strategies as needed.
• **Long-Term Financial Planning:** Define and plan for specific long-term financial goals such as retirement, buying a home, or other major life events, allocating investments accordingly.

Enter financial details... | Send

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Thank you